

timing annotations. The training set includes 212 sessions (~105 hours), and the evaluation set includes 8 sessions (~4 hours), with average speaker overlap ratios of 42.3% and 34.2%, respectively. Lastly, an additional test set contains 20 sessions (~10 hours). All speakers are native Mandarin speakers engaged in natural discussions on various topics such as business, education, and healthcare. This dataset offers a realistic and challenging benchmark for multi-speaker speech recognition in Mandarin meeting scenarios, with rich acoustic variability, high overlap rates, and dialogue structures.

In addition to AliMeeting, we further fine-tune and evaluate our method on the dataset provided by the Interspeech 2025 Multilingual Conversational Speech Language Model Challenge (MLC-SLM). This dataset consists of two-speaker real-world conversations across 11 languages: English, French, German, Spanish, Japanese, Korean, and others. The conversations cover diverse topics and are recorded in quiet indoor environments using everyday devices such as smartphones. We focus on Task 2 of the challenge, which requires joint speaker diarization and ASR without oracle segmentation or speaker labels. This makes the task significantly more challenging, as the system must autonomously determine “who spoke what and when” from unsegmented audio. Compared with the highly overlapped speech and multiple speakers in the AliMeeting dataset, the MLC-SLM dataset features low-overlap, two-speaker dialogues but introduces significant linguistic diversity. By leveraging this dataset, we aim to demonstrate the capability of our approach to generalize to multilingual scenarios.

B. Diarization System

To obtain speaker embeddings and utterance-level time boundaries required by our MS-ASR framework, we adopt the Sequence-to-Sequence Neural Diarization (S2SND) [34] system as the prior speaker diarization module. S2SND builds upon the authors’ prior work on Sequence-to-Sequence Target-Speaker VAD (Seq2Seq-TSVAD) [35], and extends it to a more general diarization framework that supports both online and offline inference without relying on clustering or permutation-invariant training. In our work, we use the S2SND-Small variant containing 16.56 million parameters. Following the original training strategy described in the paper, we train this model on the AliMeeting and MLC-SLM datasets. The generated offline diarization outputs are used to extract (i) per-speaker embeddings and (ii) sentence-level start/end timestamps. These outputs are further used as enrollment triplets in our MS-ASR system.

C. Multi-Speaker ASR Training

1) *Model Structure*: Our experimental setup employs Qwen2.5-3B [36] as the foundation model for fine-tuning, enhanced with a multi-modal architecture for speech processing. The audio pipeline utilizes the frozen encoder from Whisper-large-v3-turbo [33] as our primary speech feature extractor. At the same time, speaker representation is handled by a ResNet34 [37] network that generates both frame-

level embeddings and utterance-level embeddings for offline scenarios. The multi-modal adapter integrates a hierarchical structure consisting of three 1D convolutional layers (with kernel size 3 and progressively increasing strides of 1, 2, and 2) followed by three transformer layers featuring a 640-dimensional attention mechanism, culminating in a linear projection layer that aligns the feature space with the LLM’s input dimensions. For efficient parameter adaptation, we implement Low-Rank Adaptation (LoRA) with a rank of 8 and alpha of 16, maintaining the frozen state of all speech components throughout training to ensure stable acoustic feature extraction while enabling effective language model adaptation.

2) *Training Data*: The training data consists of two complementary sources: simulated and real conversational data. For simulated data, we generate multi-speaker mixtures by randomly combining utterances from different speakers, inserting natural silence intervals, and controlling overlap patterns. The real data pipeline extracts continuous conversational segments while preserving authentic speaker dynamics and overlap structures. Both methods employ probabilistic augmentation through embedding replacement ($p_{replace} = 0.05$), embedding dropout ($p_{drop} = 0.1$), and triplet shuffling ($p_{shuffle} = 0.2$). We implement controls including maximum segment duration (30s), per-speaker utterance limits, and variable-length sampling windows. The utterance-wised speaker embeddings are randomly selected during training and mean-pooled during inference.

3) *Configuration*: Following the findings of [38], we implement a carefully designed two-stage training strategy to optimize resource efficiency. The initial phase focuses exclusively on training the multi-modal adapter and associated triplet linear layers, while the subsequent stage introduces LoRA weights for supervised fine-tuning. Our hardware configuration utilizes 8 NVIDIA RTX A6000 GPUs with a per-device batch size of 2 and gradient accumulation steps of 4, effectively creating a larger aggregated batch size of 64. Given the limited number of training epochs, overfitting concerns are mitigated without requiring audio data augmentation. For evaluation, we employ distinct chunking strategies: the AliMeeting benchmark uses a segment limit of 10 with 4 segments per speaker, while the MLC-SLM evaluation adopts a configuration of 8 total segments with 6 segments per speaker, ensuring optimal performance for each specific task.

D. Evaluation Metric

For the speaker diarization, we report the Diarization Error Rate (DER). DER is computed as the sum of speaker confusion, missed speech, and false alarms divided by the total reference time, reflecting the diarization system’s ability to accurately detect “who spoke when.”

For the multi-speaker ASR, we first use constrained permutation Word Error Rate (cpWER) as the metric. cpWER accounts for the permutation ambiguity between speaker labels during evaluation by considering all possible alignments between system hypotheses and reference transcriptions. It is commonly used in settings where reference speaker identities

TABLE I
CPWER (%) OF OUR PROPOSED SYSTEM AND EXISTING METHODS ON THE ALIMEETING EVAL AND TEST SETS.

Method	Eval	Test
Cascaded SA-ASR		
FD-SOT [39]	41.0	41.2
WD-SOT [39]	36.0	37.1
CASA-ASR [40]	31.8	34.7
SA-Paraformer [41]	36.2	38.6
+interCTC with f&i-speaker	32.5	34.8
Ours	31.6	35.1

are known, but the system may produce transcriptions in an arbitrary speaker order.

To further evaluate the temporal accuracy of our system, we also report time-constrained permutation WER (tcpWER) [29]. This metric extends cpWER by incorporating temporal alignment constraints between predicted and reference utterances, ensuring that the content and timing are jointly correct. Specifically, tcpWER penalizes misaligned transcriptions even if the word sequence is correct, making it more suitable for speaker-attributed ASR tasks where timing precision is essential.

IV. RESULTS

To support our MS-ASR system, we first evaluate the performance of the prior speaker diarization module. Following the official evaluation protocol of the AliMeeting benchmark, which adopts Oracle voice activity detection (VAD) and a 0.25-second collar tolerance, our S2SND-based diarization system achieves a DER of 4.96% on the Eval set and 3.77% on the Test set. These outputs are used in all subsequent experiments on the AliMeeting dataset. For the MLC-SLM Challenge, which does not provide Oracle VAD and disallows any collar tolerance, our diarization module yields a DER of 14.27% on the Dev set. The ground truth annotations for the Test set remain unavailable, so no direct DER evaluation is accessible. These outputs are used for all experiments involving the MLC-SLM dataset.

Table I presents our proposed system’s cpWER (%) performance compared with several state-of-the-art multi-speaker ASR baselines on the AliMeeting Eval and Test sets. While these referenced works use the speaker-dependent Character Error Rate (SD-CER) to emphasize evaluation at the recording level with consistent speaker label permutation, this setting is now aligned with the current cpWER protocol. Therefore, we treat cpWER and SD-CER as equivalent in this context. Our system achieves the lowest cpWER on the Eval set (31.6%), outperforming the strong CASA-ASR [40] and interCTC-based Paraformer variants [41]. On the Test set, our system achieves 35.1%, slightly higher than CASA-ASR (34.7%) but still competitive.

It is worth noting that unlike the other existing systems, which are optimized solely for speech recognition accuracy, our model is designed to generate speaker-attributed transcrip-

TABLE II
TCPWER (%) OF OUR PROPOSED SYSTEM ON THE ALIMEETING EVAL AND TEST SETS BY NUMBER OF SPEAKERS.

Num. Spks	Eval	Test
2	14.94	13.57
3	31.58	29.73
4	41.59	52.67
Overall	32.17	36.36

tions with accurate utterance-level timestamps jointly. Therefore, a marginal difference in cpWER is acceptable, given our system’s richer and more structured output. These results validate the effectiveness of our diarization-aware MS-ASR framework. Specifically, the use of designed triplet enrollment enables precise utterance segmentation per speaker, while the LLM-based decoder allows joint decoding of multiple speakers with contextual modeling. This architecture is well-suited for real-world meeting transcription scenarios requiring transcript accuracy and timing alignment.

Table II reports the performance of our system on the AliMeeting Eval and Test sets using the time-constrained permutation WER (tcpWER), which simultaneously evaluates transcription accuracy and temporal alignment. Under the evaluation metric setting, a prediction is penalized even if its word content is correct as long as the predicted utterance time deviates from the reference by more than 5 seconds. This makes tcpWER a stricter and more comprehensive metric than conventional WER or cpWER. To our knowledge, our work is the first to report tcpWER results on the AliMeeting dataset. As shown in the table, the overall tcpWER is 32.17% on the Eval set and 36.36% on the Test set, reflecting our system’s ability to produce not only accurate transcriptions but also reliable utterance-level timestamps.

We further break down the results by the number of speakers in each recording. As expected, the tcpWER increases with speaker count: from 14.94% / 13.57% (2 speakers) to 31.58% / 29.73% (3 speakers), and further to 41.59% / 52.67% (4 speakers) on the Eval/Test sets. This trend reflects the increased challenge of speaker-attributed transcription under higher overlap and more frequent speaker turns. Nonetheless, our proposed system maintains reasonable performance even in the most challenging 4-speaker scenarios, demonstrating its robustness to multi-party scenarios. These results validate our design choice of combining speaker embeddings with speaking-time information to anchor utterances precisely. The tcpWER evaluation further highlights our system’s capability to meet real-world demands for timestamped and speaker-attributed transcription in long-form multi-speaker conversations.

Table III presents the tcpWER (%) of our proposed system compared to the official baseline on the dataset of MLC-SLM Challenge - Task 2, evaluated on both the Dev and Test sets. As the challenge requires, tcpWER is also computed with a 5-second tolerance window, jointly evaluating both transcription

accuracy and temporal alignment. On the Dev set, our system achieves substantial improvements over the baseline across all 15 evaluated languages. The average tcpWER is reduced from 76.12% to 24.95%, demonstrating the effectiveness of our diarization-aware MS-ASR pipeline. The improvement is particularly notable for low-resource and morphologically rich languages such as Portuguese (from 118.84% to 37.35%), French (from 96.04% to 34.74%), and German (from 86.74% to 30.38%). These results suggest that our model can generalize to diverse language patterns, even under challenging multilingual scenarios.

Moreover, the English language is evaluated across five regional variants according to the challenge protocol. Our model consistently lowers tcpWER across all English subsets, achieving reductions of more than 50% absolute in most cases, with the lowest error observed for English-Australian (14.40%) and English-Indian (16.25%). On the Test set, where only the average tcpWER is available through the official evaluation server, our model also significantly outperforms the baseline, reducing the score from 60.39% to 20.44%. This result further confirms the robustness and transferability of our approach in real-world multilingual diarization and ASR tasks.

V. CONCLUSIONS

This work presents a novel semi-end-to-end framework for multi-speaker ASR that integrates diarization-aware inputs with large language models (LLMs). Our method introduces a triplet-enrollment instruction design, combining speaker embeddings and utterance-level timing information to guide the transcription process. Unlike previous TS-ASR approaches that rely solely on speaker identity or time-range cues, our framework enables precise disambiguation of overlapping utterances, supporting the joint decoding of multiple speakers with accurate temporal alignment.

Extensive experiments on the AliMeeting dataset show that our system achieves state-of-the-art cpWER on the evaluation set and delivers competitive performance on the test set. More significantly, it produces high-quality utterance-level timestamps, as evidenced by strong tcpWER results — a metric we report for the first time on this benchmark. Additionally, experiments on the MLC-SLM Challenge demonstrate the broad generalizability of our method, with substantial tcpWER reductions across a wide range of languages.

In future work, we plan to investigate the streaming inference strategies for real-time deployment and expand the system’s applicability to downstream tasks such as meeting summarization and speaker-centric question answering.

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²<https://github.com/mubingshen/MLC-SLM-Baseline/tree/main>

TABLE III
PERFORMANCE COMPARISON (TCPWER %) ON THE MLC-SLM DEV AND TEST SETS BY DIFFERENT LANGUAGES.

Language	Official Baseline ²		Ours	
	Dev	Test	Dev	Test
English-American	53.73	-	23.01	-
English-Australian	52.63	-	14.40	-
English-British	71.92	-	18.69	-
English-Filipino	50.37	-	18.14	-
English-Indian	70.72	-	16.25	-
French	96.04	-	34.74	-
German	86.74	-	30.38	-
Italian	83.31	-	19.90	-
Japanese	71.30	-	36.29	-
Korean	59.55	-	27.04	-
Portuguese	118.84	-	37.35	-
Russian	69.21	-	23.20	-
Spanish	75.61	-	23.17	-
Thai	83.56	-	20.93	-
Vietnamese	82.80	-	29.76	-
Overall	76.12	60.39	24.95	20.44

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